

Evaluating adaptive graph fusion for short-range weather forecasting with conditional local convolution

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ABSTRACT

I evaluate an adaptive graph-fusion (AGF) input extension to the Conditional Local Convolution Recurrent Network (CLCRN) on four coarse-grid WeatherBench-derived tasks. The module gates a graph-aggregated projection against a node-wise residual projection of the same pre-CLConv input. Forty primary models (two variants, four tasks and five seeds) and six mean-fusion ablations are trained for 100 epochs in FP32. Training, validation and testing consistently include finite zero-valued targets; checkpoint selection uses validation MAE alone. Across five training seeds per variant, AGF reduces relative-humidity MAE by 2.38% and RMSE by 3.29% (Benjamini–Hochberg adjusted $q = 0.00604$ and 0.000653 , respectively, across eight primary tests). The other six comparisons do not establish a difference at $q < 0.05$. Temperature and wind have higher AGF point estimates for both metrics; cloud-cover results depend on the metric. Latitude-weighted scores, forecast-lead errors, component comparisons and sampled missing-input diagnostics provide supplementary evidence. The findings concern training-seed variability on one existing chronological split and support a task-specific humidity improvement, without establishing competitive skill against large-scale global weather models.

1 Introduction

Numerical weather prediction integrates physical governing equations and remains central to operational forecasting¹. WeatherBench established a shared setting for evaluating data-driven forecasts². The ERA5 reanalysis provides a widely used source of atmospheric training and verification fields³. Learning on these fields requires choices about spatial geometry, interactions between variables, and temporal evolution. Cubed-sphere convolutional forecasting illustrates one approach to representing the globe⁴. FourCastNet instead uses adaptive Fourier neural operators for high-resolution global prediction⁵. Pangu-Weather develops three-dimensional neural networks for medium-range forecasts⁶. GraphCast demonstrates graph-based prediction of global atmospheric states⁷. These studies motivate learned spatial representations, but their different resolutions, variables, and forecast horizons preclude direct comparisons with a small graph-recurrent benchmark.

Other weather models broaden the role of learning beyond a single deterministic forecasting architecture. ClimaX investigates pretraining across weather and climate datasets⁸. Spherical Fourier neural operators incorporate the geometry of the sphere into operator learning⁹. Neural general circulation models combine a dynamical core with learned components¹⁰. GenCast addresses probabilistic medium-range prediction¹¹. Smaller application-specific studies also address distinct needs: neural calibration of ECMWF outputs is studied by Frnda et al.¹². Zanchi et al. examine local microclimate prediction from global climate data¹³. Guo et al. investigate monthly climate prediction with convolutional and recurrent networks¹⁴. Alves et al. study refinement of numerical wind forecasts in space and time¹⁵. Their tasks differ from the 12-hour setting considered here; they provide application context rather than numerical baselines.

This study builds on the Conditional Local Convolution Recurrent Network (CLCRN), introduced by Lin et al.¹⁶. CLCRN and its Conditional Local Convolution (CLConv) operator are contributions of that prior work. CLConv uses geographic information to construct location-dependent kernels. The present data occupy a regular 32×64 latitude–longitude grid embedded on the sphere, rather than an irregular sampling grid. Figure 1 illustrates the geometric motivation; Supplementary Fig. S1 summarizes the controlled comparison. I investigate whether augmenting this geometry-conditioned backbone with a learned graph improves four short-range forecasting tasks under a matched training protocol.

1.1 Spatial representations and graph forecasting

Graph operators provide several ways to aggregate information beyond a rectangular image grid. Chebyshev spectral filtering constructs localized graph convolutions¹⁷. The graph convolutional network of Kipf and Welling

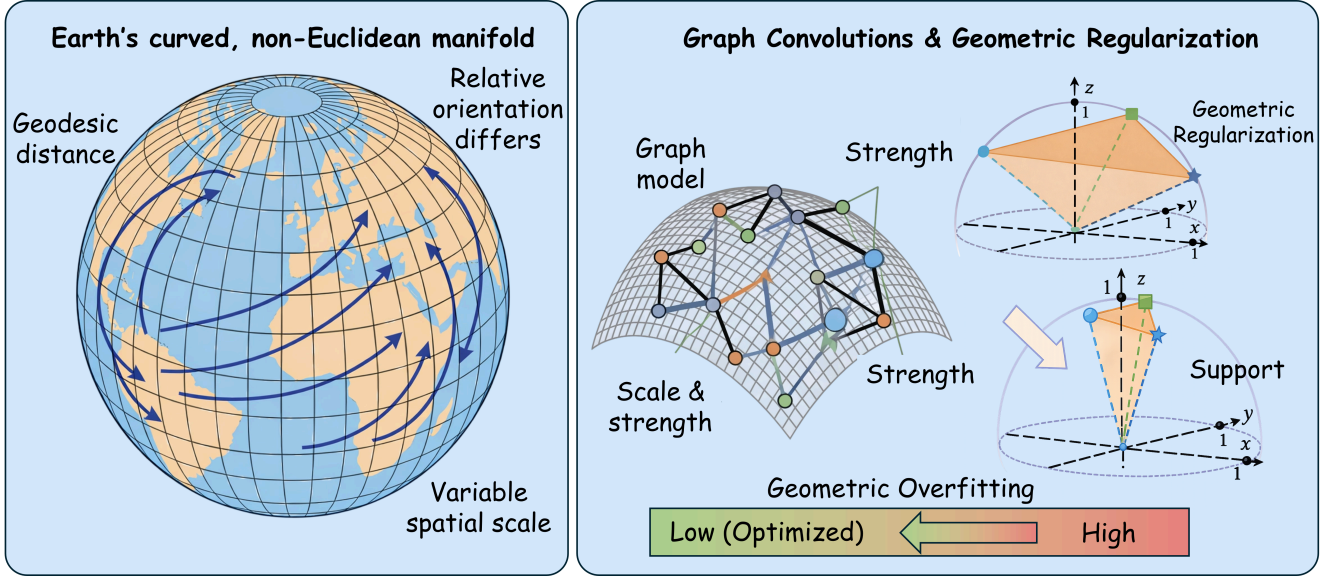


Figure 1. Geometric motivation for spatial weather modelling. The left panel illustrates variation in geodesic distance and relative orientation on the sphere. The right panel schematically depicts graph convolution and geometry-sensitive weighting. Surfaces, node positions, vectors and the labelled overfitting gradient are conceptual illustrations; they are not measured forecasts or quantitative evidence of improved regularization.

simplifies neighborhood propagation¹⁸. GraphSAGE learns aggregators that can be applied inductively to new nodes¹⁹. Graph attention assigns learned weights to neighboring features²⁰. For geometric data, PointNet demonstrates permutation-invariant processing of point sets²¹. Dynamic Graph CNN constructs feature-space neighborhoods for point-cloud processing²². Gauge-equivariant convolution provides a different, explicitly geometric treatment of signals on manifolds²³. These approaches distinguish representation, neighborhood selection, and geometric constraints; their point-cloud or classification results do not establish weather-forecasting skill. Combining graph operators with temporal dynamics has been studied extensively in forecasting, as organized in the survey by Jin et al.²⁴. Graph convolutional recurrent networks insert graph filtering into recurrent sequence models²⁵. DCRNN models directed spatial dependence using diffusion convolution²⁶. STGCN combines spatial graph convolution with temporal convolution²⁷. T-GCN couples graph convolution to gated recurrent updates²⁸. ASTGCN adds spatial and temporal attention to graph-convolutional forecasting²⁹. GMAN develops a graph multi-attention architecture³⁰. Continuous-time graph dynamics are explored by STGODE³¹. TGNN4I addresses irregular observation times and partially observed graphs³². That irregular-time setting differs from the regularly spaced hourly inputs used here, but clarifies why temporal sampling assumptions should accompany an architectural description.

1.2 Learned dependencies and fusion

Learning spatial relations from data is established practice. Graph WaveNet uses an adaptive dependency matrix alongside temporal convolution³³. AGCRN combines adaptive graph learning with node-specific parameterization³⁴. MTGNN learns directed relations for multivariate time-series forecasting³⁵. Discrete graph-structure learning treats the dependency structure itself as a learned forecasting component³⁶. DSTAGNN models dynamic spatial-temporal dependencies in traffic networks³⁷. Graph state-space models investigate latent relational structure within probabilistic temporal dynamics³⁸. Spectral temporal graph networks offer a frequency-domain formulation of multivariate forecasting³⁹. These methods motivate evaluating the benefit of a learned topology separately from the temporal backbone.

Related work also highlights design choices beyond adjacency alone. Scalable spatiotemporal graph networks use precomputed representations to reduce training cost⁴⁰. DiffSTG uses denoising diffusion for probabilistic graph forecasting⁴¹. PDFormer models propagation delays and long-range spatial dependencies⁴². STAEformer emphasizes the role of adaptive spatiotemporal embeddings in a transformer⁴³. Recent traffic studies examine multi-timescale graph representations⁴⁴. Adaptive spatiotemporal dynamic graph convolution is investigated by Zhiguo Xiao et al.⁴⁵. Hongbo Xiao et al. combine adaptive spatiotemporal attention with road-topology represen-

tations⁴⁶. These traffic applications supply methodological context, while the present experiments test a specific extension on meteorological data.

The adaptive-graph and fusion design here is inspired by ASTTN, which combines geographical and learned-graph representations⁴⁷. The distinct ASTTGN architecture studies adaptive spatiotemporal transformer graph modelling for traffic measured by loop detectors⁴⁸. The implemented AGF module in this study gates a graph-aggregated projection against a node-wise residual projection of the same pre-CLConv input. It does not fuse a post-CLConv representation with a graph branch, and it does not implement ASTTN’s Gumbel graph masking or spatial-temporal multi-head attention. Node embeddings, learned adjacency, sparsification, and gating are therefore treated as established ingredients rather than individually novel contributions.

1.3 Temporal modelling and the scope of this study

Sequence architectures provide additional context for deciding what a controlled graph-fusion experiment can establish. The transformer introduced attention-based sequence modelling⁴⁹. Informer investigates efficient attention for long-sequence forecasting⁵⁰. Autoformer combines decomposition with an autocorrelation mechanism⁵¹. PatchTST uses patches to represent time series⁵². Crossformer models dependencies across dimensions⁵³. iTransformer represents variables as tokens to learn multivariate dependencies⁵⁴. TimesNet maps temporal variation into a two-dimensional representation⁵⁵. N-BEATS uses basis-expansion blocks for forecasting⁵⁶. Mamba develops selective state-space sequence modelling⁵⁷. Results showing the competitiveness of simple linear forecasting models further motivate careful controls when attributing gains to architecture⁵⁸. These models are not rerun here; comparisons among their published scores would confound datasets, horizons, and evaluation conventions. The contribution of this study is an AGF input extension to the published CLCRN framework and a controlled empirical evaluation. The comparison uses identical data, training schedules, and budgets for the two local variants, five training seeds, uncertainty estimates, and multiplicity correction. Forty primary models and six mean-fusion ablations connect the reported results to selected checkpoints and figure data. Component comparisons and diagnostic analyses accompany the task-specific findings, including negative or inconclusive results. This scope supports a test of incremental graph fusion rather than a claim of a new forecasting architecture or competitive skill against large-scale global weather models. The following Methods–Results–Discussion structure specifies the implementation and evaluation conventions before interpreting the numerical findings.

2 Methods

2.1 Data, chronological split, and normalization

I use the four meteorological tasks adopted by CLCRN¹⁶. Their WeatherBench-derived targets² are 2-m temperature (K), relative humidity (percentage points), 10-m wind components (u, v ; m s^{-1}), and total cloud cover (fraction). The two wind components are predicted together; the other tasks are trained separately. Hourly data from 1 January 2010 through 31 December 2018 are supplied on a regular 32×64 latitude–longitude grid ($N = 2,048$). Each example contains 12 hourly inputs followed by 12 hourly targets. Windows start 24 hours apart, yielding 3,286 examples. Chronological, unshuffled train/validation/test partitions contain 2,300/329/657 examples. Thus successive examples do not share a 24-hour input–target window. For each physical variable (and separately for each wind component), the normalization mean and standard deviation are computed over the training input arrays across examples, input times, and nodes. The same statistics transform inputs and targets. In the frozen implementation, these statistics are computed from the original training-input arrays. Inputs and targets are then cast to float32 before in-place subtraction of the training mean and division by the training standard deviation. This float32-before-standardization order is used for training, validation and test inference and is recorded in the processing-version manifest. Forecast errors are calculated after inverse standardization.

2.2 Backbone and implementation-aligned AGF encoder

The CLConv–GRU encoder–decoder is based on the public CLCRN implementation. CLConv uses coordinates, geodesic distance, and orientation information to condition local convolution. These operators and the recurrent update are retained; AGF augments the encoder input rather than replacing CLConv. Both local variants use trainable node embeddings, whereas the upstream reference implementation registers its node embeddings as a fixed buffer. Thus the control is a matched local CLCRN implementation retaining the published CLConv–GRU operators, rather than an unchanged reproduction of every upstream implementation choice. During training, the autoregressive decoder uses curriculum-based teacher forcing. At evaluation, it consumes its own previous predictions.

Let $\bar{\mathbf{x}}_{bti} \in \mathbb{R}^C$ be the standardized input at example b , time t , and node i . A linear weather projection and a learned node embedding, each of dimension 8, form the AGF input

$$\mathbf{F}_{bti} = \phi_x(\bar{\mathbf{x}}_{bti}) \parallel \mathbf{e}_i^{\text{node}} \in \mathbb{R}^{16}. \quad (1)$$

Source and destination embeddings $\mathbf{E}_s, \mathbf{E}_t \in \mathbb{R}^{N \times 10}$ define a directed, row-normalized adjacency,

$$A_{ij} = \frac{\exp(\mathbf{e}_i^s \top \mathbf{e}_j^t)}{\sum_{\ell=1}^N \exp(\mathbf{e}_i^s \top \mathbf{e}_\ell^t)}. \quad (2)$$

This adjacency uses the same softmax of source–destination embedding products as ASTTN (Eq. 8)⁴⁷; the graph aggregation and fusion implementation used here is specified below. This graph is parameter dependent but does not change with the input weather sample. The implementation retains the self-edge and the largest $k-1$ non-self entries, with $k = 4$, then renormalizes:

$$\mathcal{T}_k(i) = \{i\} \cup \text{top}_{k-1}\{A_{ij} : j \neq i\}, \quad \tilde{A}_{ij} = \frac{A_{ij}}{\sum_{\ell \in \mathcal{T}_k(i)} A_{i\ell}} \quad (j \in \mathcal{T}_k(i)). \quad (3)$$

Here top_{k-1} returns indices. Selection does not exclude geographically nearby nodes, so the learned edges are potentially long-range but are not guaranteed to be nonlocal. Aggregation, projections, and fusion follow the source-code order:

$$\mathbf{m}_{bti} = \sum_{j \in \mathcal{T}_k(i)} \tilde{A}_{ij} \mathbf{F}_{btj}, \quad (4)$$

$$\mathbf{q}_{bti} = \mathbf{W}_m \mathbf{m}_{bti} + \mathbf{b}_m, \quad \mathbf{r}_{bti} = \mathbf{W}_r \mathbf{F}_{bti} + \mathbf{b}_r, \quad (5)$$

$$\mathbf{g}_{bti} = \sigma(\mathbf{W}_g[\mathbf{r}_{bti} \parallel \mathbf{q}_{bti}] + \mathbf{b}_g), \quad (6)$$

$$\mathbf{z}_{bti} = \mathbf{g}_{bti} \odot \mathbf{q}_{bti} + (1 - \mathbf{g}_{bti}) \odot \mathbf{r}_{bti} \in \mathbb{R}^8. \quad (7)$$

Both branches act on the same pre-CLConv input. In particular, $\mathbf{r}$ is a node-wise residual linear projection, not a geometry-conditioned CLConv representation. The encoder receives $[\mathbf{F} \parallel \bar{\mathbf{x}} \parallel \mathbf{z}]$; the control receives $[\mathbf{F} \parallel \bar{\mathbf{x}}]$. The corresponding input widths are 25 versus 17 for the scalar tasks and 26 versus 18 for wind. There is no calendar input, periodic encoding, or multi-head attention in the primary configuration. Figure 2 shows the corrected computation.

2.3 Matched new training and checkpoint selection

Both models use two recurrent layers with 32 hidden units, 8-dimensional feature and node embeddings, one CLConv propagation view, and 25 geographic neighbours. Adam uses learning rate 0.01 and $\epsilon = 0.001$, batch size 32, gradient-norm clipping at 5, and MultiStepLR milestones $\{10, 20, 30, 40, 50\}$ with factor 0.95. Every run completes 100 epochs; early stopping is disabled. The curriculum-learning decay parameter is 2,000. All primary and ablation runs use FP32, with automatic mixed precision and TF32 disabled. Training and validation MAE are evaluated after inverse standardization, including every finite target, including physically valid zero values. Testing uses the same zero-inclusive support. Every epoch is saved atomically; the earliest epoch attaining the minimum full-validation MAE is selected, without consulting the test set. Checkpoints retain optimizer, scheduler, random-number-generator and scaler states to permit interruption recovery. All trainable parameters are materialized before optimizer construction, and optimizer membership and gradient coverage are checked explicitly.

The prior archive used a zero-exclusive objective. Its scores are superseded by the newly trained results reported here. The correction therefore applies to optimization and validation selection, rather than solely rescoring historical checkpoints. Code is frozen before training, and per-run manifests record source hashes, configuration, seed, precision and environment.

The training environment recorded on 5 October 2026 is Windows 11, Python 3.13.7 and PyTorch 2.10.0 with CUDA 12.8. The device is an NVIDIA GeForce RTX 5090 Laptop GPU (driver 610.88; 24,463 MiB reported memory). The manifest additionally records NumPy 2.4.2, SciPy 1.17.0 and pandas 3.0.0. These are the recorded training settings; rendering the revised figures and compiling the manuscript do not require GPU training.

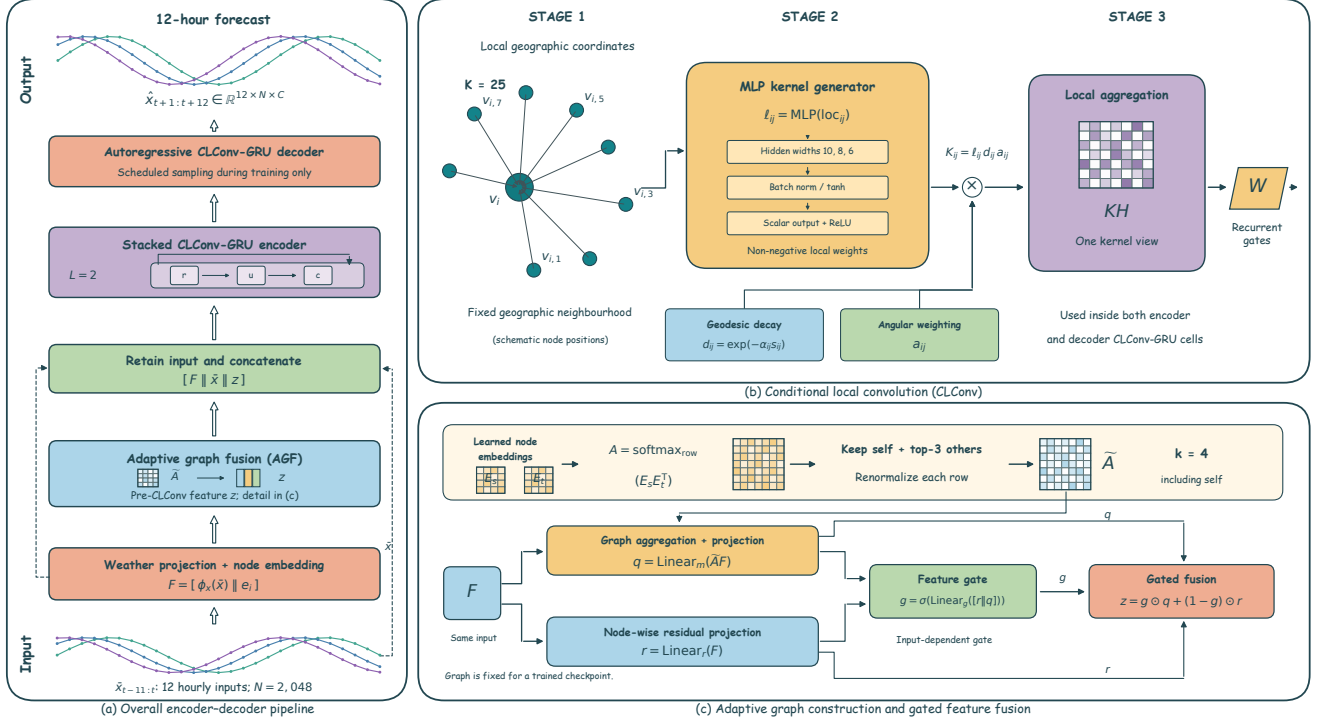


Figure 2. Implementation-aligned CLCRN-AGF architecture. (a) Weather projections and node embeddings form F . The AGF feature z is concatenated with F and standardized weather before the CLConv-GRU encoder and autoregressive decoder. (b) CLConv combines a coordinate-conditioned local kernel, geodesic decay and angular weighting over fixed geographic neighbourhoods. These experiments use one kernel view. (c) The learned adjacency retains self and the top $k - 1$ other neighbours, with $k = 4$, followed by renormalization. Both the graph-aggregated projection q and the node-wise residual projection r act on the same pre-CLConv input; the gate produces $z = g \odot q + (1 - g) \odot r$. Each Linear operator includes its learned bias and acts on the feature axis. For a fixed checkpoint the adjacency is input-independent, while the gate is input-dependent. The control omits z and retains the weather and node features. Input traces, node layouts and matrix cells are schematic.

2.4 Test metrics and replicate-level inference

The primary MAE and RMSE pool all finite targets over test examples, the 12 forecast times, nodes, and output components with equal weight:

$$\text{MAE} = \frac{1}{M} \sum_{a=1}^M |\hat{y}_a - y_a|, \quad \text{RMSE} = \sqrt{\frac{1}{M} \sum_{a=1}^M (\hat{y}_a - y_a)^2}. \quad (8)$$

These are unweighted grid-point metrics, not area-weighted scores. I also report a latitude-weighted sensitivity analysis, replacing the unit weights with $w_i = \cos \phi_i$ (latitude ϕ_i in radians) and dividing weighted error sums by the sum of weights over valid entries. This changes the evaluation metric only; it does not change the training objective or checkpoint selection. For cumulative horizon h , errors pool steps $1, \dots, h$; exact-step scores use only step h . Persistence repeats the last input observation and is scored by the same test implementation. Each primary task has five independently trained control and five AGF replicates. Reusing seed labels does not create matched random streams, because AGF changes the parameter initialization sequence. I report means, sample standard deviations, AGF-minus-control differences, two-sided Welch 95% confidence intervals, and two-sided Welch tests. Benjamini-Hochberg correction is applied across the eight task/metric tests. The 95% intervals are marginal, not simultaneous or multiplicity-adjusted intervals; the adjusted q values govern the primary significance statements. A two-sided exact permutation test of the difference in means provides a sensitivity check under the null of exchangeable group labels, over the $\binom{10}{5} = 252$ group allocations. This replaces the previous

paired sign-test interpretation. With only five seeds per group, these intervals characterize training variability on one fixed split; they do not quantify uncertainty across climates, datasets, or test periods. A non-significant result is not evidence of equivalence. With five replicates per group, distributional assumptions are difficult to assess; the exact permutation check is also conditional on exchangeability and does not remove this small-sample limitation.

2.5 Provenance and supplementary experiments

The fixed experiment manifest contains 40 newly trained primary models: control and AGF for four tasks and seeds 2021–2025. Six additional newly trained mean-fusion models cover temperature and humidity with seeds 2023–2025. Every model completes the same 100-epoch FP32 protocol. The corresponding control and gated-AGF ablation rows reuse exactly those three seeds from the primary experiment; they are not separately selected favourable runs. Without AGF, the entire input augmentation is removed. Without a learned gate, the fusion is $\mathbf{z} = (\mathbf{q} + \mathbf{r})/2$.

All accuracy tables, horizon plots, weighted scores and gate diagnostics are regenerated from this common set of new checkpoints. Missing-input diagnostics use a documented shared subset. Historical tuning trials and short screening runs are not used as substitutes for these replicates. Source and checkpoint hashes, seed-level CSV files, validation-selection records and figure metadata are retained with the analysis.

The external CLCRN reference means are traceable to Table 2 of the published AAAI paper¹⁶. Temperature and wind scores retain their displayed units; humidity errors are multiplied by 10 and cloud-cover errors divided by 10 to match the units used here. These historical values were not recomputed with the present evaluator. Weather-Bench’s standard evaluator uses cosine-latitude weights, whereas the released CLCRN evaluator uses unweighted means and masks zero targets. I therefore state both conventions explicitly without assuming that a historical reference shares the corrected zero-inclusive protocol. The source audit and unit-conversion table are retained in the accompanying reproducibility materials under `historical_context`; the primary table contains only matched scores.

2.6 Computational scope

The AGF implementation constructs a dense $N \times N$ similarity matrix before top- k selection, so graph construction has quadratic intermediate memory despite sparse aggregation. CLConv also materializes dense kernel operators. The recorded trainable parameter counts are 39,268 for the scalar control and 80,892 for scalar AGF; wind uses 39,437 and 81,061, respectively. The inherited inference microbenchmark is identified separately as a measurement of historical weights. Its timings are not relabelled as measurements of the newly trained checkpoints.

AI-assisted preparation. AI tools assisted language editing, code review, analysis scripting and figure preparation. Their use is described in the Generative AI and AI-Assisted Technologies Disclosure section.

3 Results

3.1 Primary five-seed comparison

The primary results use the same zero-inclusive, spatially unweighted evaluator for all 40 newly trained primary checkpoints. Table 1 reports the five-seed comparison. Only relative-humidity MAE and RMSE meet the primary threshold after Benjamini–Hochberg correction across eight tests. The remaining comparisons are inconclusive at $q < 0.05$.

For relative humidity, mean MAE decreases from 4.648 to 4.537 percentage points (2.38%; $q = 0.00604$), and mean RMSE from 7.248 to 7.010 percentage points (3.29%; $q = 0.000653$). Temperature and wind have higher point estimates for both metrics with AGF, while cloud-cover MAE is slightly higher and RMSE slightly lower. The marginal 95% intervals for all six of these differences include zero; the experiment does not establish either improvement or equivalence for these tasks.

Persistence repeats the final input observation. Table 2 and Fig. 3 compare it with the two trained variants on the same test split and evaluation support. Both trained variants have lower mean MAE and RMSE than persistence for all four tasks. The heatmap ranks the displayed means within each task and metric; its colours do not encode statistical significance or a common error scale across variables. Published CLCRN scores remain source context rather than matched baselines. The Supplementary Information section “Persistence on the same split” and Supplementary Fig. S3 provide the corresponding contextual results.

Figure 4 shows the individual MAE replicates and their sample standard deviations. In temperature, AGF has a larger MAE sample SD than the control (0.107 versus 0.060 K). The replicate spread is part of the reported uncertainty and does not support a general stability benefit. Seed labels are not treated as paired observations.

Table 1. Five-replicate comparison of CLCRN-AGF and the schedule-matched CLCRN control. Values are mean $\pm$ sample standard deviation across seeds 2021–2025. Differences are CLCRN-AGF minus control; negative values favour CLCRN-AGF. Confidence intervals and two-sided p -values use Welch’s unequal-variance approximation.

Task	Metric	Control (mean $\pm$ s)	CLCRN-AGF (mean $\pm$ s)	Difference [95% CI]	p_{Welch}
Temperature	MAE	1.258 $\pm$ 0.060	1.291 $\pm$ 0.107	+0.033 [−0.100, +0.166]	0.5701
	RMSE	2.101 $\pm$ 0.045	2.118 $\pm$ 0.129	+0.017 [−0.141, +0.174]	0.7942
Relative humidity	MAE	4.648 $\pm$ 0.017	4.537 $\pm$ 0.039	−0.111 [−0.158, −0.063]	0.0015
	RMSE	7.248 $\pm$ 0.043	7.010 $\pm$ 0.055	−0.239 [−0.311, −0.166]	8.17×10^{-5}
Surface wind	MAE	1.242 $\pm$ 0.011	1.256 $\pm$ 0.037	+0.014 [−0.032, +0.059]	0.4699
	RMSE	1.972 $\pm$ 0.023	2.016 $\pm$ 0.093	+0.043 [−0.071, +0.158]	0.3657
Cloud cover	MAE	0.1479 $\pm$ 0.0017	0.1484 $\pm$ 0.0020	+0.0005 [−0.0022, +0.0031]	0.7077
	RMSE	0.2500 $\pm$ 0.0017	0.2490 $\pm$ 0.0028	−0.0010 [−0.0045, +0.0025]	0.5103

All scores include valid zero targets and use equal spatial weights over the full 12-hour forecast and 657 test examples. Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Benjamini–Hochberg correction across the eight comparisons retains only the two humidity differences at $q < 0.05$ (MAE $q = 0.00604$; RMSE $q = 0.000653$).

Table 2. Contextual test results on the same test split. Persistence is deterministic; control and CLCRN-AGF are means across five seeds. All methods use the same zero-inclusive, spatially unweighted evaluation over the full 12-hour forecast. Lower is better.

Method	Temperature		Relative humidity		Surface wind		Cloud cover	
	MAE	RMSE	MAE	RMSE	MAE	RMSE	MAE	RMSE
Persistence	1.421	2.700	6.425	10.447	1.537	2.535	0.1620	0.2725
CLCRN control (5 seeds)	1.258	2.101	4.648	7.248	1.242	1.972	0.1479	0.2500
CLCRN-AGF (5 seeds)	1.291	2.118	4.537	7.010	1.256	2.016	0.1484	0.2490

Errors are in K for temperature, percentage points for relative humidity, m s^{-1} for surface wind, and fractions for cloud cover. Variation across model seeds is reported in Table 1.

Current-split results under one evaluation protocol

	Temperature MAE (K)	Temperature RMSE (K)	Humidity MAE (pp)	Humidity RMSE (pp)	Wind MAE (m s^{-1})	Wind RMSE (m s^{-1})	Cloud cover MAE (fraction)	Cloud cover RMSE (fraction)	
Persistence	1.421 #3	2.700 #3	6.425 #3	10.447 #3	1.537 #3	2.535 #3	0.1620 #3	0.2725 #3	Relative value within each column Best Mid-range Worst
Control (n = 5)	1.258 #1	2.101 #1	4.648 #2	7.248 #2	1.242 #1	1.972 #1	0.1479 #1	0.2500 #2	
CLCRN-AGF (n = 5)	1.291 #2	2.118 #2	4.537 #1	7.010 #1	1.256 #2	2.016 #2	0.1484 #2	0.2490 #1	

Cell colours use separate min-max scales per column; numbers show errors in physical units.
Lower is better. Ranks are descriptive; pp = percentage points. Persistence is deterministic.

Figure 3. Cumulative 12-hour MAE and RMSE for persistence, the CLCRN control and CLCRN-AGF on the same test split. Trained-model values are five-seed means; persistence is deterministic. Values match Table 2 and use zero-inclusive, spatially unweighted evaluation. Colours are normalized independently within each task/metric column; the cell values retain physical units. Rankings are descriptive. Temperature is in K, relative humidity in percentage points (pp), wind in m s^{-1} and cloud cover in fraction.

Five-replicate comparison of the adaptive graph-fusion extension

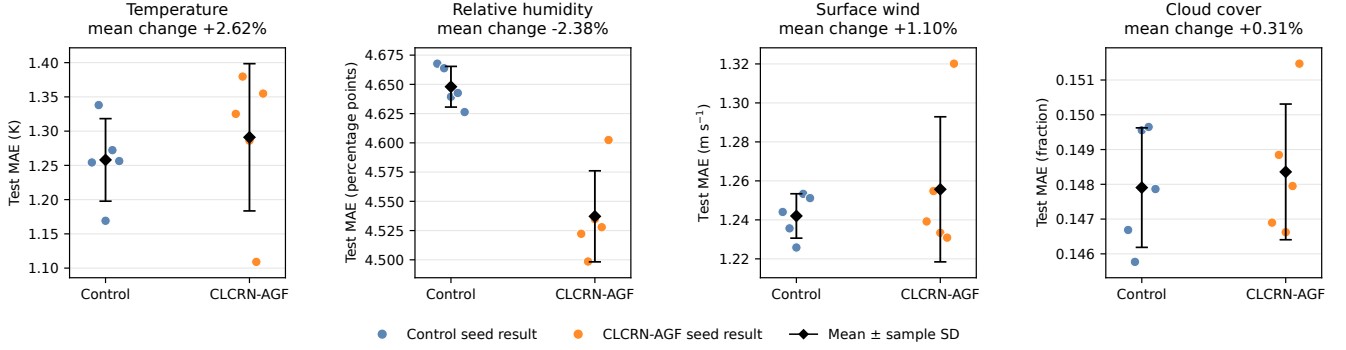


Figure 4. Cumulative 12-hour test MAE for the same 40 primary checkpoints as Table 1. Coloured points are five independently trained replicates per variant; black diamonds and error bars show the mean plus or minus sample SD. Panel annotations give $100(\overline{\text{MAE}}_{\text{AGF}}/\overline{\text{MAE}}_{\text{control}} - 1)$; negative values favour AGF. Panels have task-specific units and focused vertical scales. The error bars describe training-seed variability, not confidence intervals.

3.2 Statistical and spatial-weighting sensitivity

The exact permutation analysis supports the same two relative-humidity comparisons after separate BH correction across its eight tests ($q = 0.03175$ for both MAE and RMSE). The other six adjusted values range from 0.771 to 0.778. Welch tests remain the primary analysis; the permutation results are a sensitivity check with the exchangeability qualification in Methods. Supplementary Fig. S2 displays the primary mean differences and marginal Welch intervals on a relative scale.

Latitude weighting changes only the evaluation metric. The Supplementary Information section “Latitude-weighted sensitivity” reports cosine-latitude-weighted scores from the same new checkpoints. Relative humidity again has lower mean MAE and RMSE (2.73% and 3.86%, respectively); its adjusted values are 0.01076 and 0.000871. These eight Welch tests form a separate sensitivity family, not additional primary discoveries.

3.3 Forecast horizon and matched-protocol summary

Figure 5 shows exact-lead MAE, cumulative MAE and exact-lead RMSE for the five AGF checkpoints per task. These are the AGF half of the primary experiment. Exact-lead errors describe an individual forecast hour, whereas cumulative errors pool all hours up to that lead. Consequently, the cumulative 12-hour MAE, rather than the exact 12-hour MAE, equals the AGF entry in Table 1. The three curves describe the AGF forecasts and do not by themselves compare AGF with the control at each lead. Supplementary Fig. S6 presents the exact-lead and cumulative MAE curves for both variants.

Figure 6 summarizes cumulative 12-hour MAE with zero-based axes. It uses exactly the means and sample SDs in Table 1 and Fig. 4. Both variants complete the same 100-epoch FP32 training protocol and use validation-only checkpoint selection. The bars therefore compare the two model variants under this shared protocol; they do not decompose the effects of changing a training schedule or compare an individual run with a replicate mean.

3.4 Precision-matched component comparison

The temperature and humidity ablations use seeds 2023–2025 for all three variants, each with 100 epochs and FP32. Control and gated AGF are the corresponding primary subset; six mean-fusion runs are newly trained under the same protocol. The means and sample standard deviations describe these three replicates. For humidity, the gated model has lower mean MAE and RMSE than mean fusion. For temperature, the ordering depends on the metric: the control has the lowest MAE, whereas mean fusion has the lowest RMSE (Table 3). These are descriptive comparisons with only three replicates per variant; they neither establish a universal gate benefit nor enter the eight-test primary family. Removing AGF removes the complete augmentation, including its residual projection, and changes parameter count, so this comparison does not isolate the learned adjacency alone.

3.5 Missing-input robustness

Both variants are evaluated on 64 deterministically spaced common test windows. Node-mask ratios 0.1, 0.2, 0.3 and 0.4 use mask seeds 11, 22 and 33, shared between variants. Masks remain fixed across input hours and sig-

Per-Step Forecast Error vs. Prediction Horizon

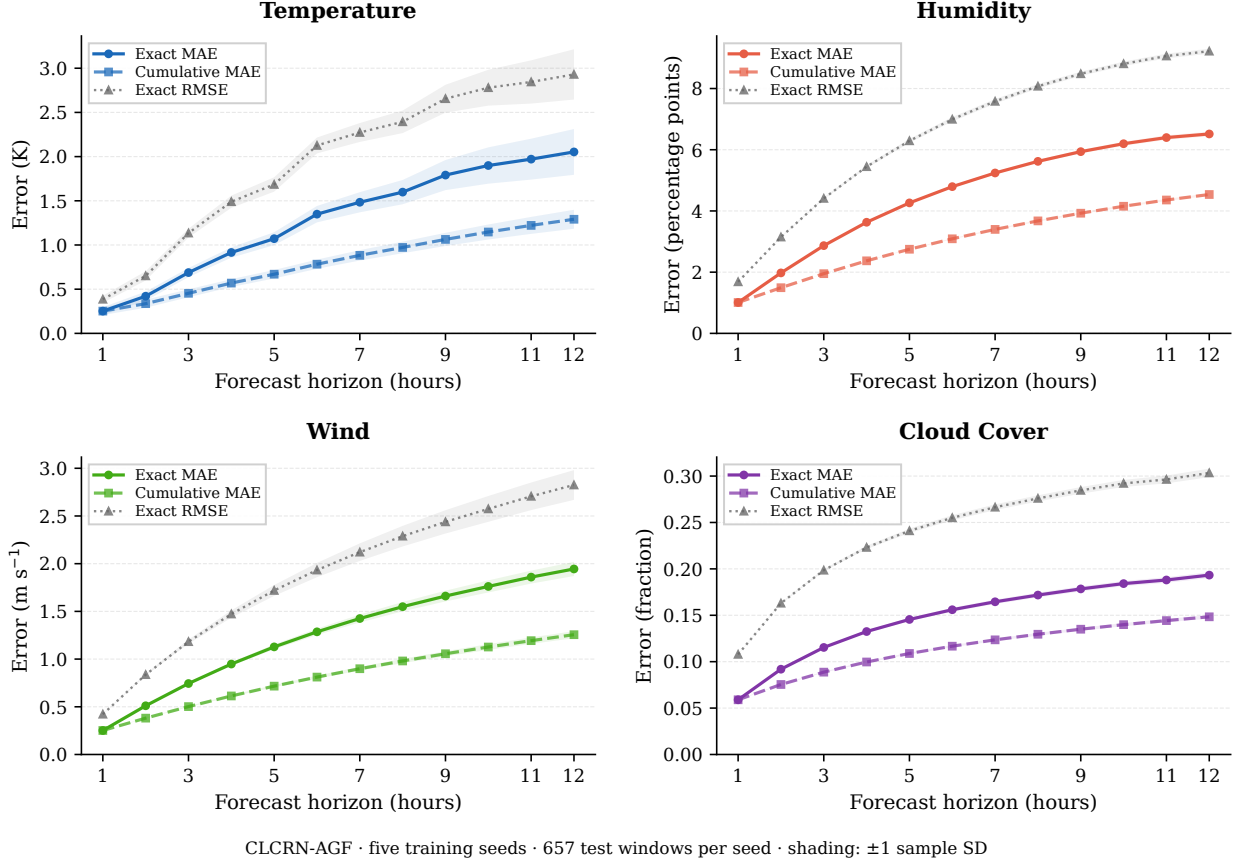


Figure 5. Forecast-horizon errors for five CLCRN-AGF checkpoints per task on all 657 test windows. Solid circles show exact-lead MAE, dashed squares show cumulative MAE over leads 1 through h , and grey triangles show exact-lead RMSE. Lines are training-seed means; shading denotes plus or minus one sample SD for each curve. The cumulative MAE at 12 hours equals the corresponding AGF mean in Table 1. Panels use independent scales and physical units. All finite targets, including zero, are scored with equal spatial weights.

CLCRN control and adaptive graph-fusion comparison

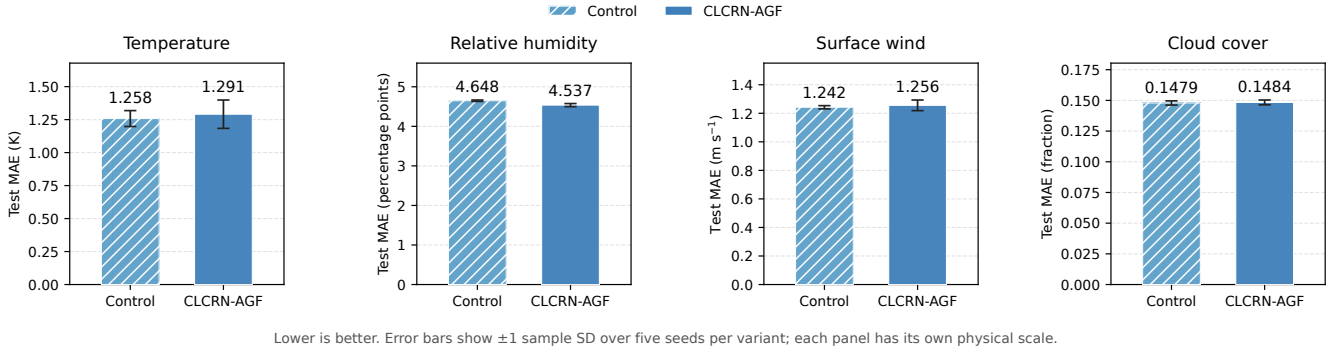


Figure 6. Cumulative 12-hour test MAE under the matched training and evaluation protocol. Bars are five-seed means and error bars are plus or minus sample SD, matching Table 1 and Fig. 4. All axes start at zero; panels have task-specific units and independent scales. This is a model comparison under a common schedule, not an estimate of a training-schedule effect.

Table 3. Component ablations with a matched 100-epoch FP32 training protocol. All entries report mean $\pm$ sample standard deviation over seeds 2023–2025 on the full 12-hour test forecast. Lower is better.

Budget	Task	Variant	MAE	RMSE
100 epochs, 3 seeds	Temperature	Control (no AGF)	1.233 ± 0.056	2.113 ± 0.043
		w/o gated fusion	1.267 ± 0.066	2.047 ± 0.103
		Full CLCRN-AGF	1.250 ± 0.127	2.057 ± 0.138
100 epochs, 3 seeds	Relative humidity	Control (no AGF)	4.636 ± 0.009	7.227 ± 0.043
		w/o gated fusion	4.599 ± 0.024	7.132 ± 0.062
		Full CLCRN-AGF	4.555 ± 0.041	7.040 ± 0.048

The control removes the entire AGF module, including its residual projection, changing the encoder input width and parameter count. The ungated variant uses a fixed equal average of the graph and residual branches. The control and full-model rows use the same seed subset from Table 1. Errors are in K for temperature and percentage points for relative humidity.

nal channels; masked standardized values are replaced by zero, corresponding to the training mean. Mask results are averaged within each training seed before summarizing five seeds. Figure 7 displays the AGF results as relative changes from each checkpoint’s clean score on those same 64 windows. Complete results for both variants are retained in the accompanying analysis data and Supplementary Fig. S5.

At 40% masking, the mean relative MAE increases for AGF are 465.9% for temperature, 74.7% for humidity, 84.7% for wind and 37.5% for cloud cover. The corresponding control increases are 474.1%, 73.0%, 85.7% and 35.5%. The descriptive ordering therefore varies across tasks. The large temperature degradation concerns this training-mean-imputation stress test and does not demonstrate behaviour under naturally occurring missingness. These sampled diagnostics neither establish full-test robustness nor provide a significance test for robustness differences.

3.6 Full-test gate diagnostics

Gate statistics retain all eight scalar channels, 2,048 nodes, 12 input hours, 657 test windows and five AGF seeds per task (Table 4; Fig. 8). This yields 645,857,280 scalar gates per task. The pooled means are 0.600 for temperature, 0.570 for humidity, 0.508 for wind and 0.475 for cloud cover. These coefficients weight the graph-aggregated projection in Eq. 7; the complement weights the residual projection of the same pre-CLConv input. The pooled activation SD describes variation across all retained activations, not a between-seed standard error.

The profiles in Fig. 8 use the recorded 100-bin histograms. Their quantile markers are approximations from the binned counts; the reported means and SDs are calculated from the full activation stream. A coefficient near one half does not imply equal information contribution, because branch magnitudes and directions also matter. Neither a gate distribution nor a learned graph alone establishes a meteorological mechanism. Supplementary Fig. S4 provides the corresponding histogram representation.

Table 4. Gate-activation statistics pooled over all five CLCRN-AGF seeds and the full test set. The gate balances a graph-aggregated projection and a residual linear projection of the same input, before CLConv. Standard deviations describe the pooled population of scalar gates.

Variable	Mean μ	Std σ	Min	Max
Temperature	0.600	0.196	0.001	1.000
Relative humidity	0.570	0.244	0.001	1.000
Surface wind	0.508	0.130	0.049	0.954
Cloud cover	0.475	0.208	0.002	0.997

Each task includes $5 \times 657 \times 12 \times 2,048 \times 8 = 645,857,280$ scalar gates (seeds, test examples, input steps, nodes and channels). Channels are retained individually; no channel averaging or test-set subsampling is applied.

3.7 Computational cost in context

Across the 46 completed runs, the sum of recorded epoch training-plus-validation durations is 56.34 GPU wall-clock hours on a serial single-GPU queue. This sum excludes initialization, final test inference, interruption/restart overhead and supplementary analysis; it is not an end-to-end campaign-duration or energy estimate.

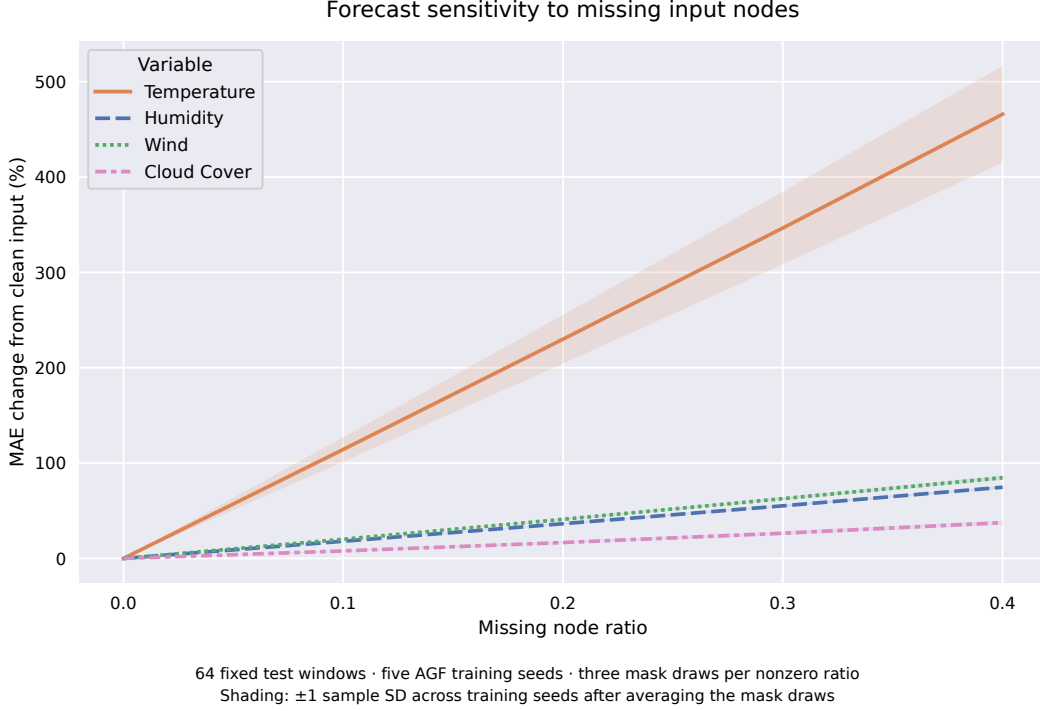


Figure 7. Missing-input-node sensitivity of five AGF checkpoints per task on the same 64 test windows at every masking ratio. The plotted change is $100(\text{MAE}_{\text{masked}}/\text{MAE}_{\text{clean}} - 1)$, using each checkpoint’s clean score on that subset. A node mask is fixed across all 12 input hours and signal channels; standardized inputs are set to zero (training-mean imputation). At each nonzero ratio, three mask seeds (11, 22, 33) are averaged within each training seed. Lines and shaded bands show the mean plus or minus sample SD across five training seeds. The clean point is zero by definition. The bands describe seed variability, not confidence intervals or significance tests; this diagnostic uses 64 of the 657 test windows.

Historical-weight inference microbenchmark. The following inherited timing evidence uses archived temperature seed-2021 weights, not the newly trained weights in the accuracy tables. It measures a fixed batch of 32 standardized windows, FP32, 12 inputs and 12 outputs, on an RTX 5090 Laptop GPU after 10 warm-up iterations and 30 synchronized repetitions. Historical control/AGF mean batch latencies were 220.53/205.60 ms (iteration SD 18.53/5.43 ms), with peak allocated memory 333.65/614.50 MiB. This sequential repeated-batch profile does not establish a speed advantage, does not measure new-weight inference, and excludes preprocessing. The historical benchmark data remain separately labelled in the supplied revision package.

Context relative to recent global weather models. The scope of CLCRN-AGF is a controlled, task-specific forecast on a 32×64 grid, using 12 hourly inputs to predict the next 12 hours. Recent global ensemble systems address a substantially broader task (Table 5). WeatherNext 3 reports improved probabilistic skill over WeatherNext 2 for most upper-level targets and surface fields in its 2024 evaluation, with some exceptions at short lead times⁵⁹. FourCastNet 3 reports skill competitive with GenCast and better than IFS-ENS in its 2020 WeatherBench 2-style evaluation⁶⁰. These reports do not establish an accuracy ranking against CLCRN-AGF: the verification years, initial conditions, spatial resolutions, variables, forecast horizons, spatial weighting and deterministic versus probabilistic scores differ. In particular, pressure-level specific humidity is not interchangeable with the relative-humidity target used here. A numerical head-to-head comparison would require harmonized data and verification.

Training cost and practical scope. The reported training resources also differ substantially. WeatherNext 3 uses about 6.7 wall-clock days per model seed; its two seeds jointly require about 4.8 TPUv4 chip-years and 8.9 TPU7x chip-years⁵⁹. FourCastNet 3 reports three stages: 78 h on 1,024 H100 GPUs, 15 h on 512 A100 GPUs, and 8 h on 256 H100 GPUs⁶⁰. These resources are reported separately from inference, and different accelerator types are not converted into a common cost unit. The single-member inference timings in Table 5 are neither full-ensemble timings nor measurements on the hardware used here. CLCRN-AGF is therefore positioned as a

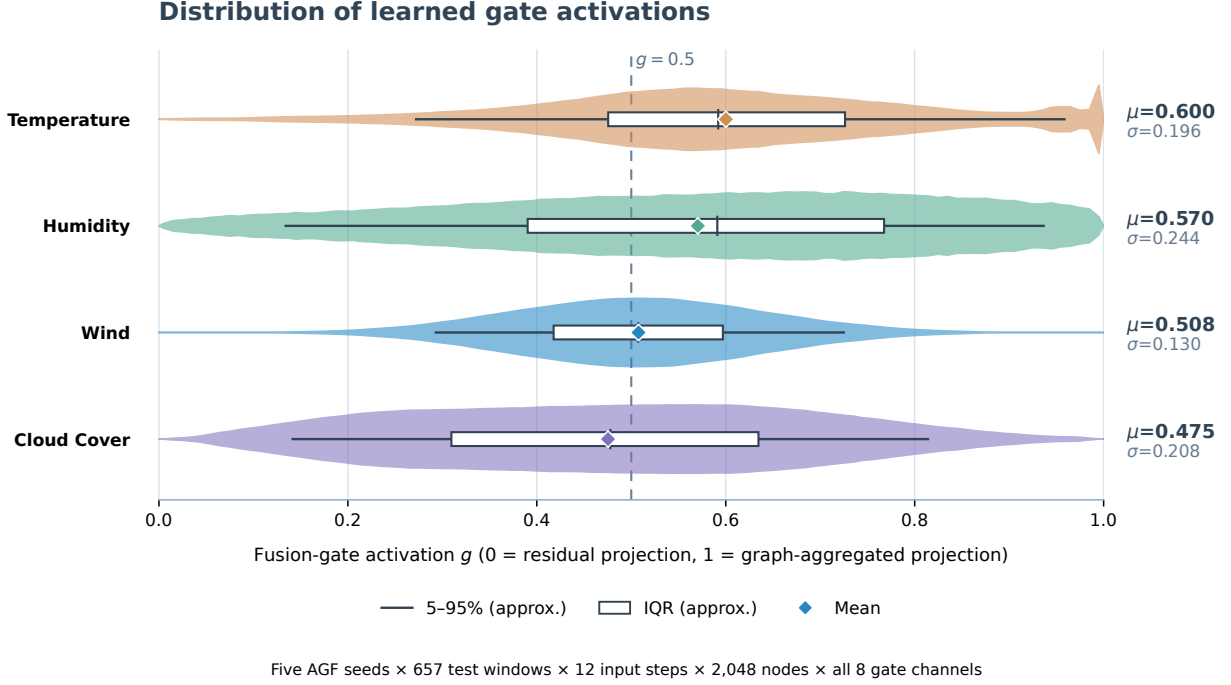


Figure 8. Scalar gate activations pooled over all five AGF checkpoints and the complete test set for each task, with all channels retained. Mirrored profiles interpolate the centres of the recorded 100-bin histograms on $[0, 1]$; widths are normalized separately within each task. White boxes, centre lines and whiskers show approximate 25th–75th, 50th and 5th–95th percentiles, obtained by linear interpolation of cumulative bin counts. Diamonds and the displayed means and population SDs use the full-stream statistics in Table 4. The dashed line marks $g = 0.5$. The gate mixes graph and residual projections of the same pre-CLConv input; the profile widths and quantiles do not measure predictive contribution.

compact setting for retraining and controlled methodological studies; this experiment does not establish competitive operational forecast skill or superior speed at matched resolution and forecast coverage.

Table 5. Context from published global-model reports, not a common-test accuracy or hardware benchmark. WeatherNext 3 specifications are taken from the official documentation⁶¹ checked on 8 October 2026. Timings are per ensemble member as reported by the respective research papers.

System	Spatial/temporal output and horizon	Predicted fields	Reported inference
CLCRN-AGF (this study)	32×64 ; hourly; 12 h	Separate temperature, relative-humidity, wind and cloud-cover tasks	Single RTX 5090 Laptop GPU; historical-weight batch profile only; no new-weight timing claim
WeatherNext 3 ⁵⁹	0.1° surface, 0.25° pressure levels, 0.05° station head; hourly surface output; 15 days for main cycles	Surface/upper-air fields, cloud, radiation, precipitation and station temperature/dewpoint	About 6.3 min per member on 4 TPUv5p chips
FourCastNet 3 ⁶⁰	0.25° ; 6-hourly; 15-day skill evaluation, rollouts assessed to 60 days	72 channels: 7 surface fields and 5 variables at 13 pressure levels	About 1 min per 15-day member on one H100 GPU

4 Discussion

The evidence supports a modest relative-humidity improvement from AGF under this fixed protocol, with lower MAE and RMSE after the eight-test correction. It does not support a general improvement across weather targets. Temperature and wind have higher mean errors with AGF, while cloud-cover MAE and RMSE move in different directions. All six marginal intervals for these tasks include zero. The larger temperature replicate spread also prevents a claim of uniformly more stable training.

All primary and mean-fusion models are newly trained with zero-inclusive training and validation, saved every epoch and selected by validation alone. The reported accuracy study therefore uses a common target support for optimization, checkpoint selection and testing. Its figures and tables use the same selected checkpoints; the horizon plot shows the AGF subset, and the missing-input analysis uses an explicitly identified common 64-window subset. These conventions permit consistent comparisons without assuming that historical scores used the same objective or evaluator. Negative and inconclusive outcomes are retained in the same tables.

The gate combines graph and residual projections of a common pre-CLConv input. Neither its learned edges nor its scalar coefficients are identified as meteorological teleconnections, moisture transport or causal contribution. Full-test activation profiles summarize the fitted gates, but their pooled spread does not establish repeatability across training seeds or explain the humidity improvement. The component comparison is precision matched, but only three seeds and two tasks are available for mean fusion. Removing the entire AGF augmentation also removes its residual branch and changes model size, so that ablation does not isolate the effect of the adjacency. Missing-input diagnostics concern sampled windows and training-mean imputation; the task-dependent changes do not support a general robustness advantage. No new season-stratified experiment is claimed.

Five independently initialized runs per variant on one chronological split give limited precision and no uncertainty assessment across different climate periods. The seed-level intervals do not treat correlated grid points or forecast hours as independent experimental replicates. The fixed test split was used in earlier development and revision work; new seeds therefore do not constitute evaluation on a newly reserved test period. The coarse grid, separate task targets and 12-hour horizon remain distinct from modern multivariate medium-range forecasting. Contemporary models require harmonized data, compute and verification before a direct accuracy ranking is meaningful. Dense graph construction limits scaling, and the historical-weight microbenchmark is not a runtime study of the newly trained models.

5 Conclusion

In this controlled experiment, AGF lowers relative-humidity MAE and RMSE by 2.38% and 3.29%, respectively, with both comparisons meeting the eight-test BH threshold. No difference is established for the other six primary comparisons. The result is a task-specific finding about training-seed variability on the existing chronological split, not evidence of general improvement or equivalence across targets. The completed 40-run primary study and six precision-matched ablations provide a traceable evaluation of an incremental AGF input augmentation. The claims follow the observed effects, marginal uncertainty intervals and multiplicity-adjusted tests; they do not assert architectural priority for the individual AGF ingredients or parity with global weather models.

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7 Author Contributions

Xinchen Geng conceived the extension, implemented the software, conducted the experiments and analysis, prepared the visualizations, and drafted, reviewed and revised the manuscript.

8 Data Availability

The ERA5-derived WeatherBench data are available from <https://mediatum.ub.tum.de/1524895>; underlying ERA5 data are available through <https://cds.climate.copernicus.eu/>. The original CLCRN preprocessed-data link is documented at <https://github.com/EDAPINENUT/CLCRN>. Third-party weather arrays are not duplicated in full in the revision package. A standardized 32-window temperature batch is retained under `historical_context` solely for the separately labelled historical inference microbenchmark. The seed-level results, checkpoint provenance and analysis inputs are archived in release

v0.3-review-reproducibility-20261008: <https://github.com/gengx-prog/CLCRN-AGF-Scientific-Reports/releases/tag/v0.3-review-reproducibility-20261008>. The current manuscript source package includes the frozen CSV/JSON inputs used for its figures and tables. The revised visual presentation uses these same experiment records. Earlier releases are retained as historical materials and are not the source of the primary accuracy results here.

9 Code Availability

The public experiment release v0.3-review-reproducibility-20261008 contains frozen training and analysis sources, the 46-run plan, dependency and environment records, and one validation-selected checkpoint for each run. It also retains complete 100-epoch training histories, run summaries, model configurations and checkpoint hashes. These selected checkpoints support re-evaluation; the release does not contain all 4,600 epoch checkpoints. Full epoch archives remain in the local experiment directory. Recovery states in each selected checkpoint pertain to that epoch, not necessarily epoch 100. Third-party weather arrays must be obtained from the cited sources. The repository is <https://github.com/gengx-prog/CLCRN-AGF-Scientific-Reports>. Scripts and frozen inputs for the figures and tables in this manuscript accompany its source under `figure_reproduction`. They reproduce the present visual presentation from the recorded metrics; model training and test inference use the separate experiment release.

10 Competing Interests

The author declares no competing interests.

11 Generative AI and AI-Assisted Technologies Disclosure

During preparation of this revision, the author used AI tools for language editing and assistance with code review, analysis scripting, and figure preparation. These tools assisted preparation of the materials; scientific interpretation and responsibility for the manuscript rest with the author.

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